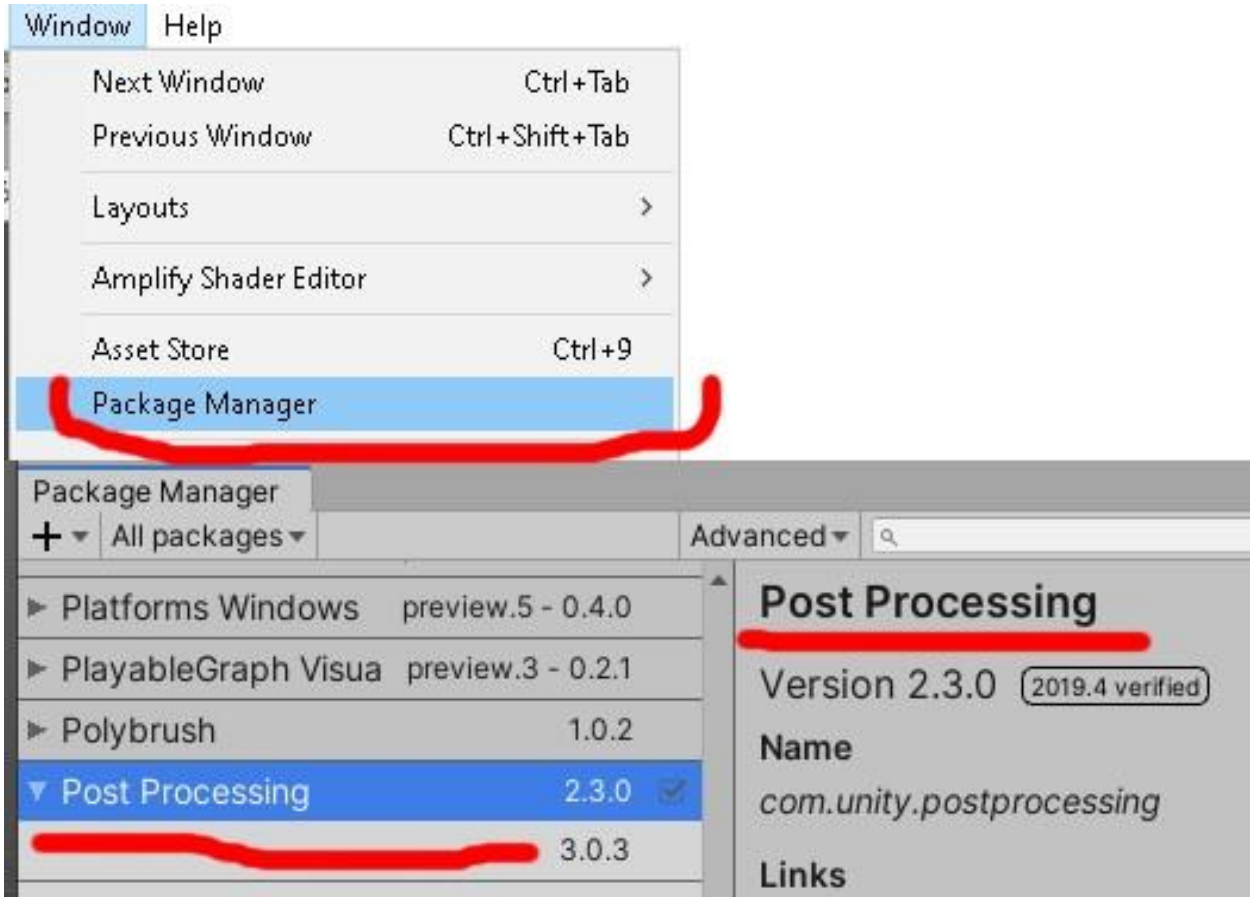
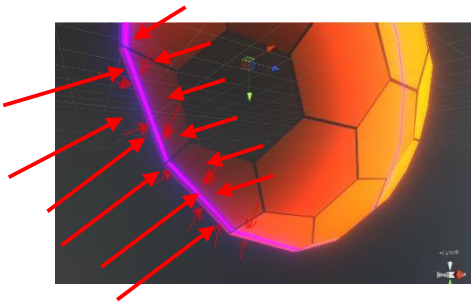
1. Post Processing.

- To make the effects work correctly, install Post Processing before using the pack.



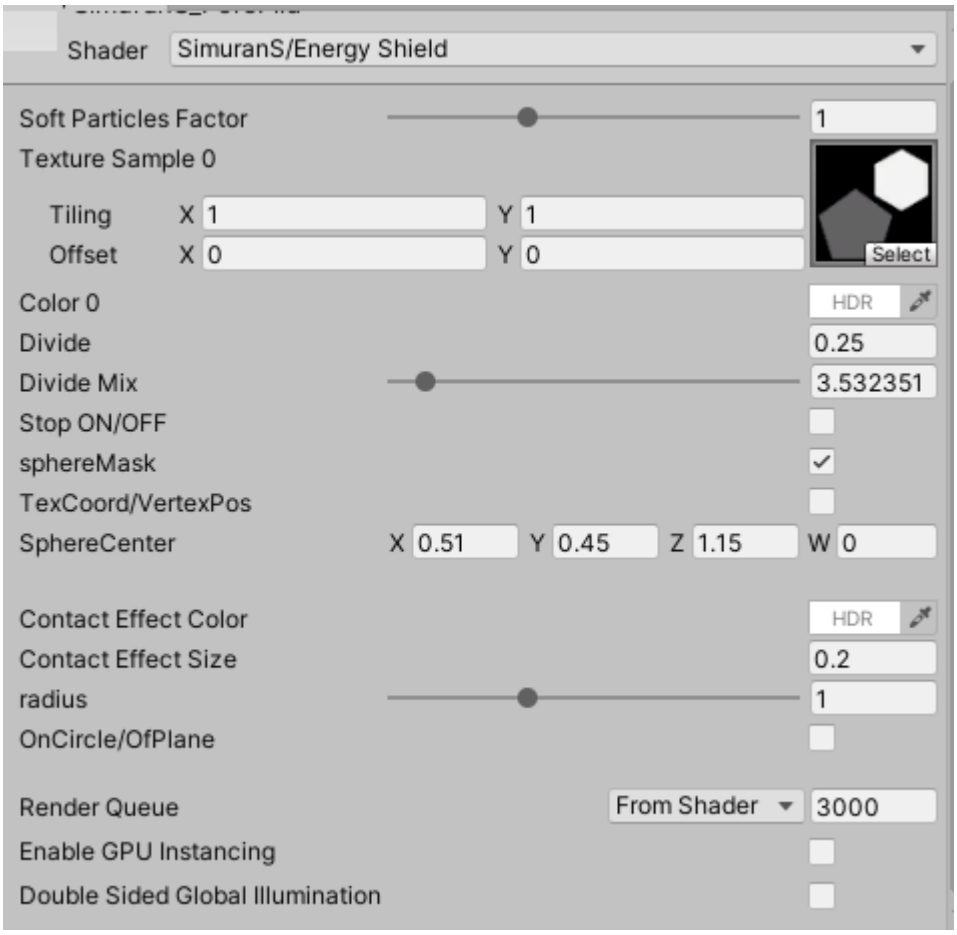
2. Managing shaders materials.

- Effects work on Build-in/ HDRP / LWRP(URP).



- If you are using HDR/LWRP(URL), set the value of Contact Effect Size = 0

2.1 Energy Shield Material Parameters.

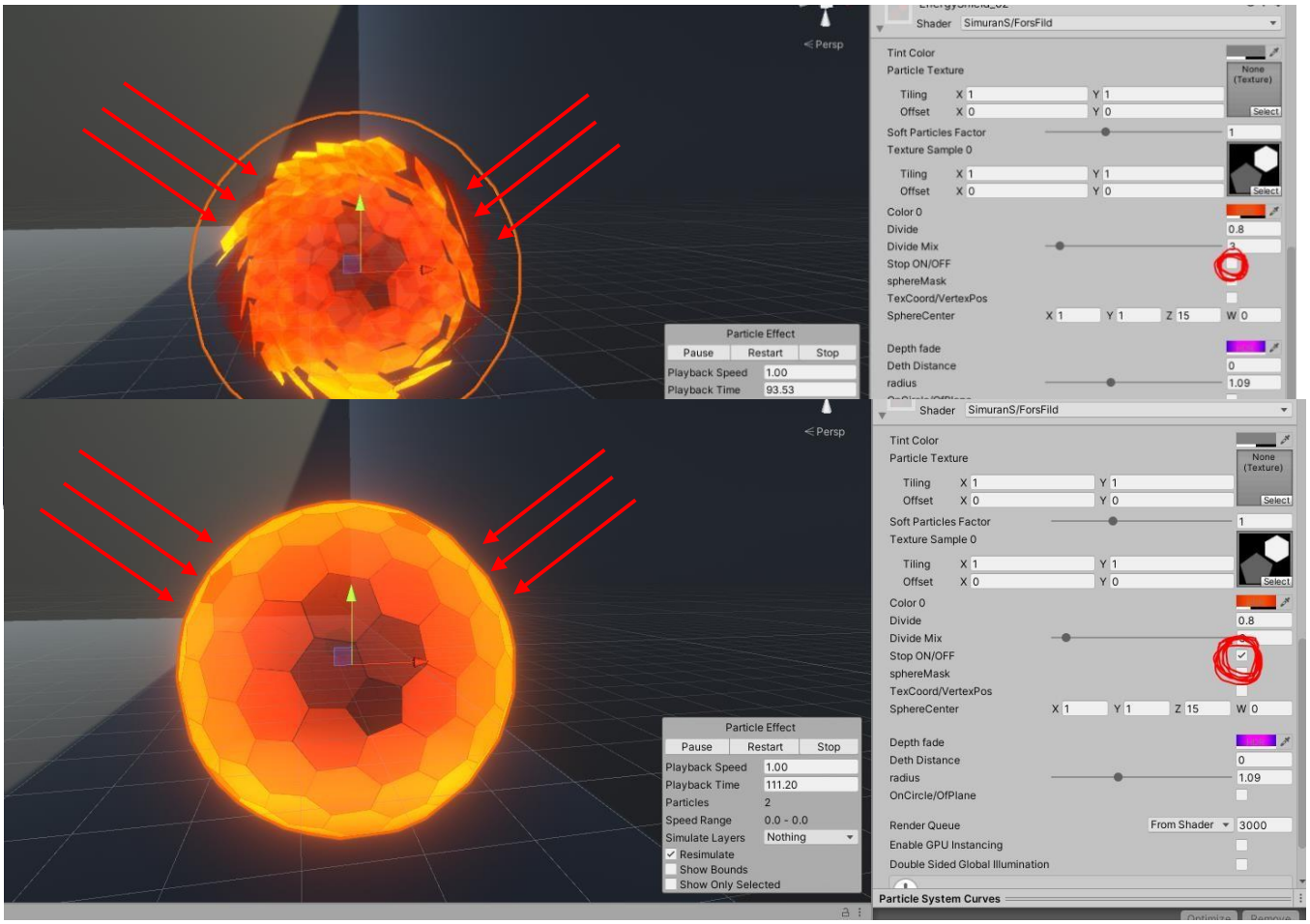


1. Texture Sample 0 – The texture that defines the main pattern of the effect.

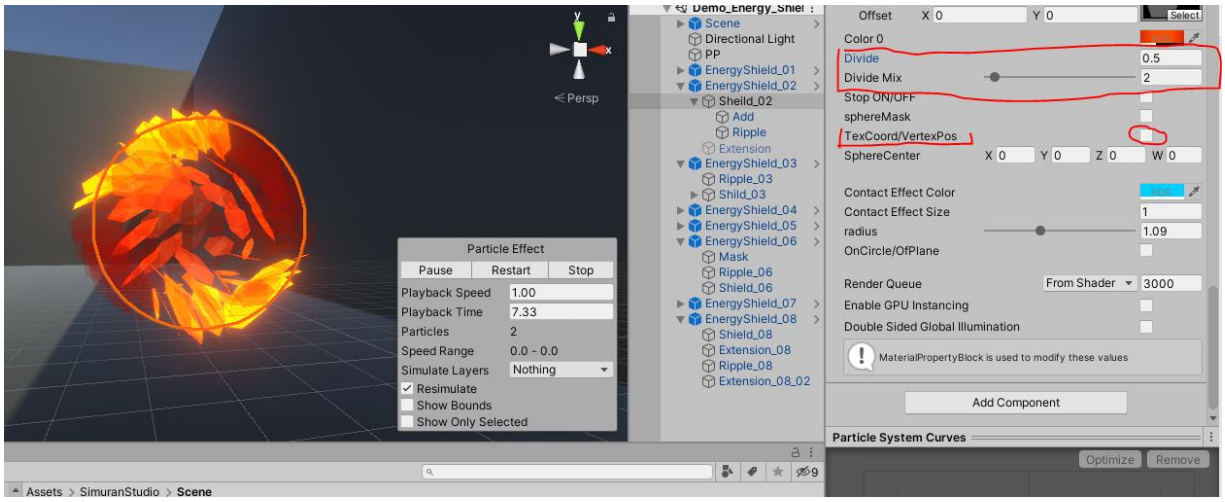
2. Color 0 – Color of texture in HDR to connect and adjust the glow.
3. Divide & Divide Mix
   3.1. Divide - the amplitude of the narrowing and expansion of the shield during pulsation.
   3.2. Divide Mix – the rate of narrowing and expansion of the shield during pulsation.

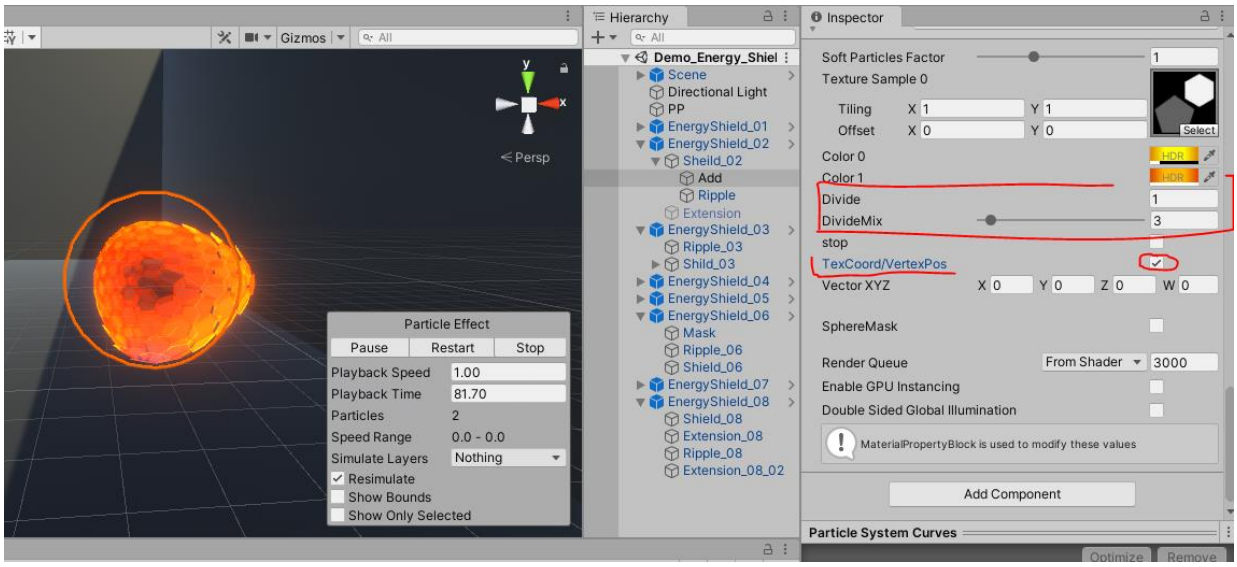
(* if any of the values of Divide & Divide Mix is 0, the pulsation effect does not work)

4. Stop ON/OFF – if true, it does not allow the shield to narrow when the pulsation is less than the value set in the Start Size variable in the Particle System.



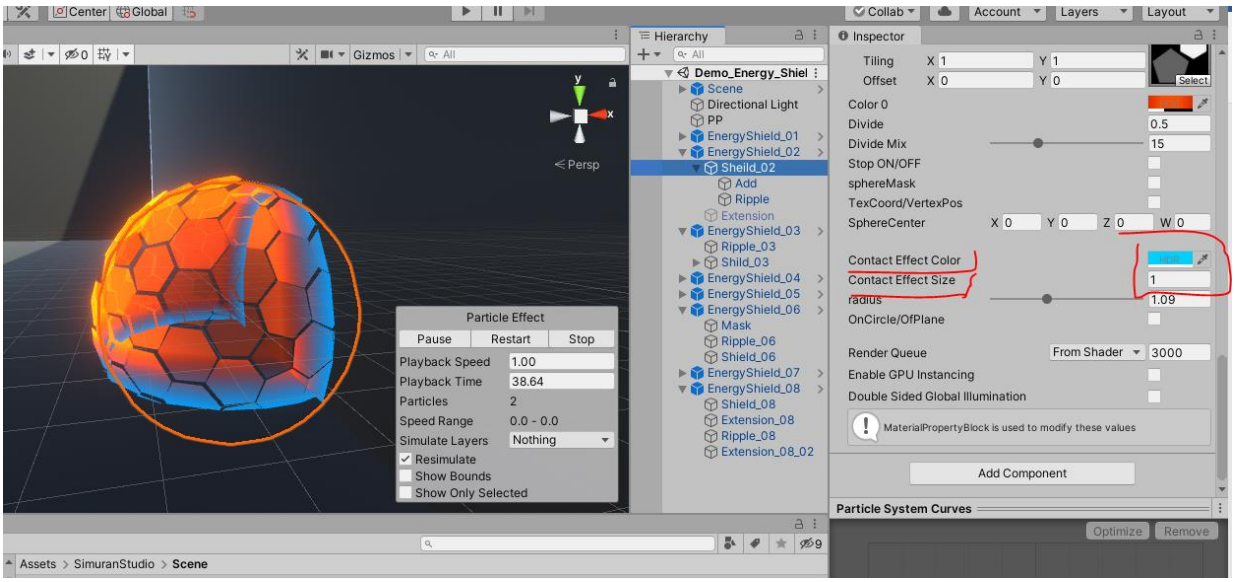
5. SphereMask – if this is true, the shield is invisible in the standby state and appears only when a projectile hit it. Else we see the shield all the time.
6. TexCoord/VertexPos – if false, the shield pulsates, retaining its original shape. Else, it randomly changes shape, like a jelly bubble.





7. SphereCenter is a service variable. Use it in your script to display impacts if you use other weapons besides the demo cannon provided in the project. (The change does not affect anything; it is just output so that the name for the scripts is visible)
8. Contact Effect Color – the color of the bright Distance Field strip on the edge of intersection with other objects.

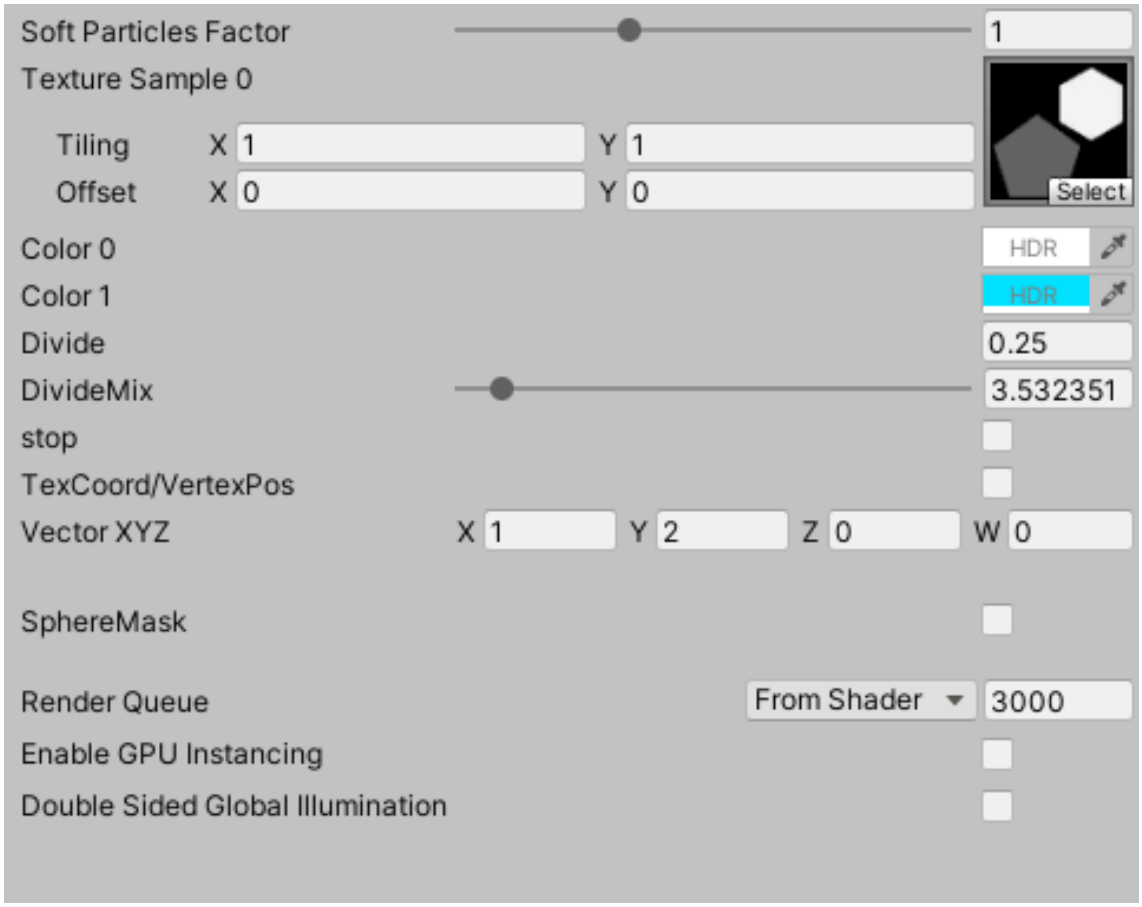
9. Contact Effect Size – the size of the bright Distance Field strip on the edge of intersection with other objects.

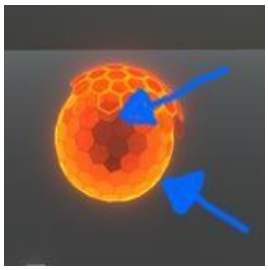


10. Radius – The radius of the spot appearing at the point of impact of the projectile on the shield.

11. OnCircle/Ofplane – You need to turn it on for three-dimensional objects, and turn it off for planes.

2.2 Fresnel(additional shader material).



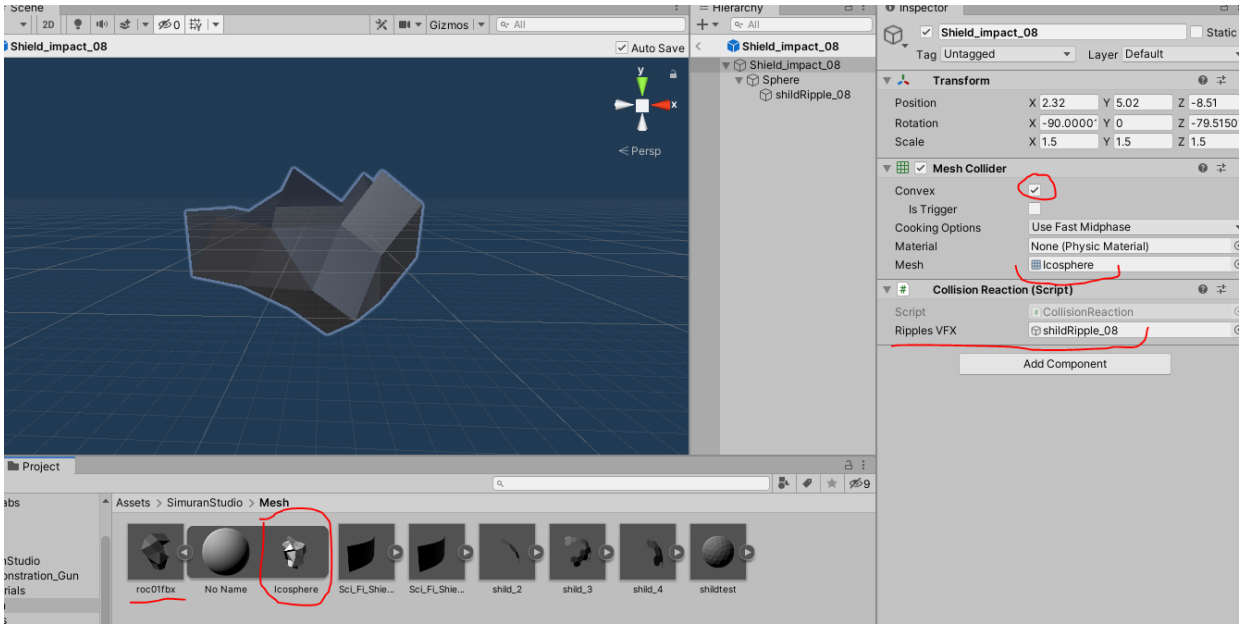


- An additional shield that creates a translucent sphere with a Fresnel refraction, which allows you to make the effect of a translucent bubble with a gradient brighter to the contour and a transparent center. It has all the same parameters and values as the first shield, with one exception: if you enable SphereMask, then by shifting the Vector XYZ variable you can configure an incomplete shield like a hemisphere, etc.

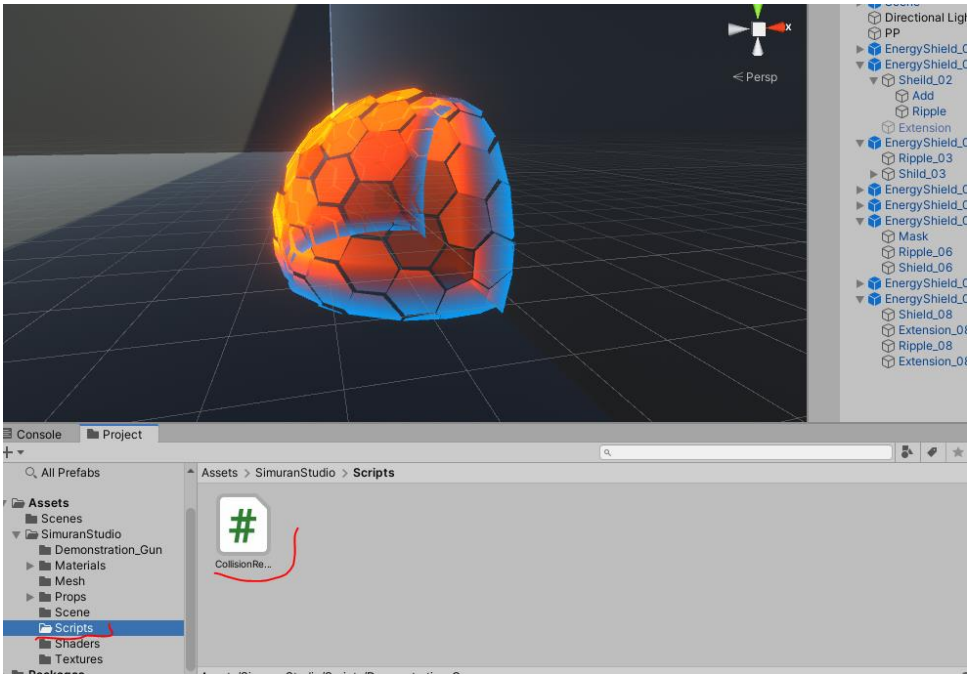
2.3 Ripple (additional impact shader material).

This allows you to make a power shield that appears when projectiles are hit, not only spherical in shape, but also to surround any mesh with it. To connect this to your mesh, perform the following actions:

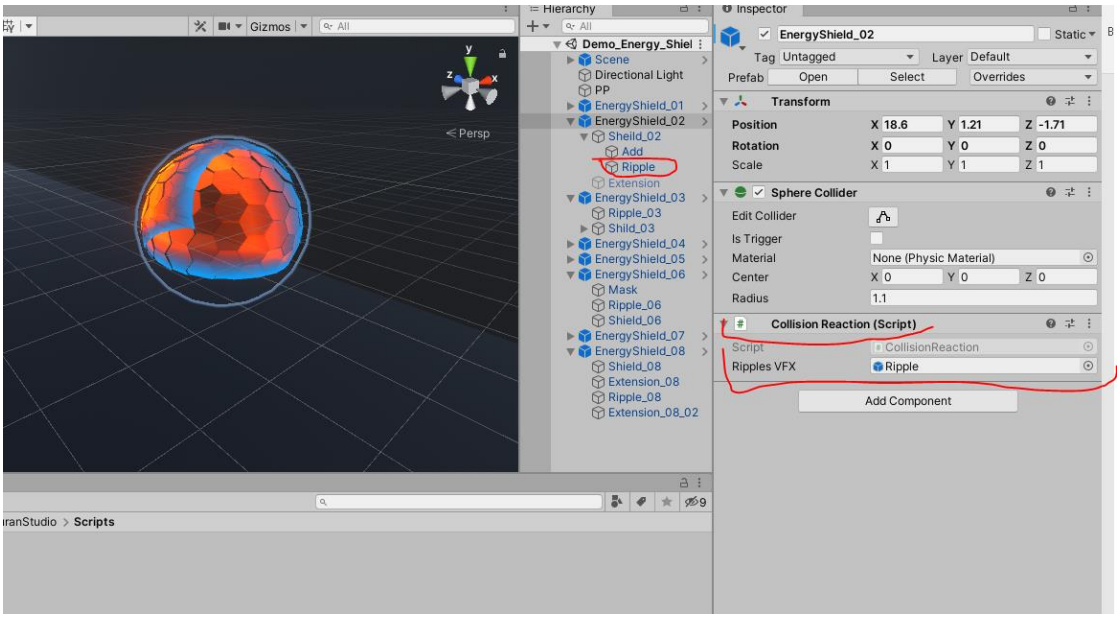
1. Select the Mesh on which you want to apply the Ripple effect.
2. Switch the Convex variable to True to display the effect correctly.



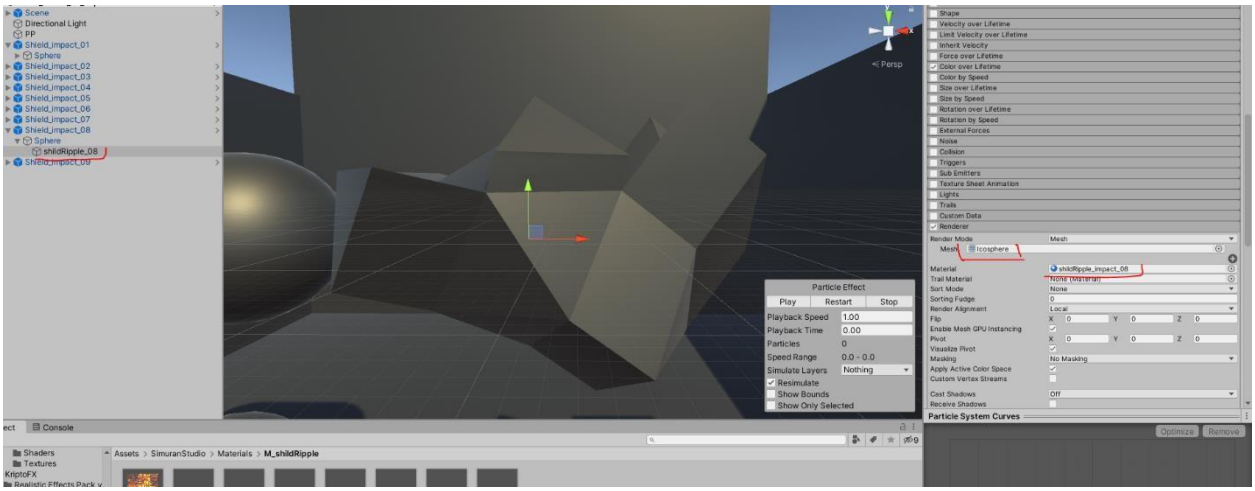
3.You need to put a CollisionReaction to connect the effect



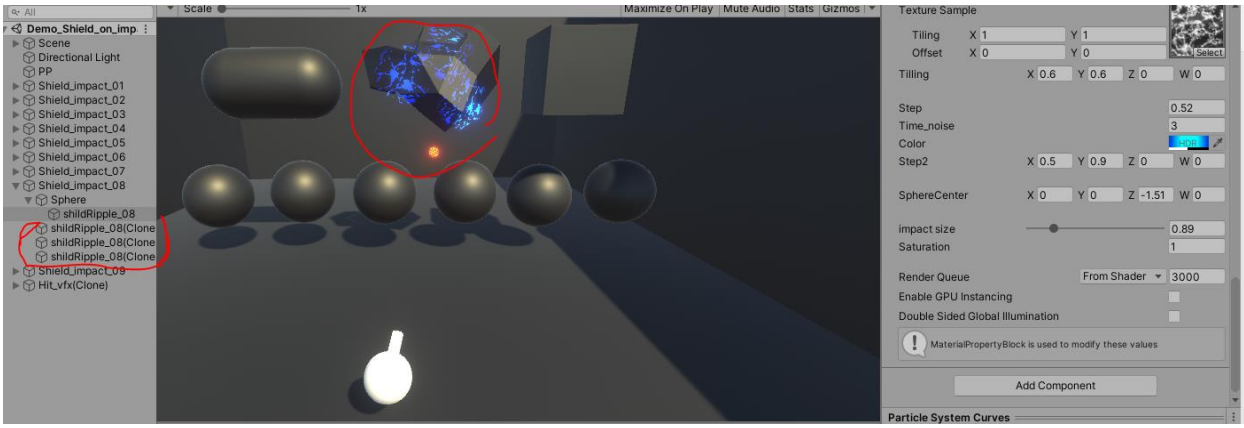
4.Ripple VFX - assign the material for this system.



5. In the partial system render, set the object you need to the Mesh variable.
6. Put the created Material from the Ripple shader into the Material.

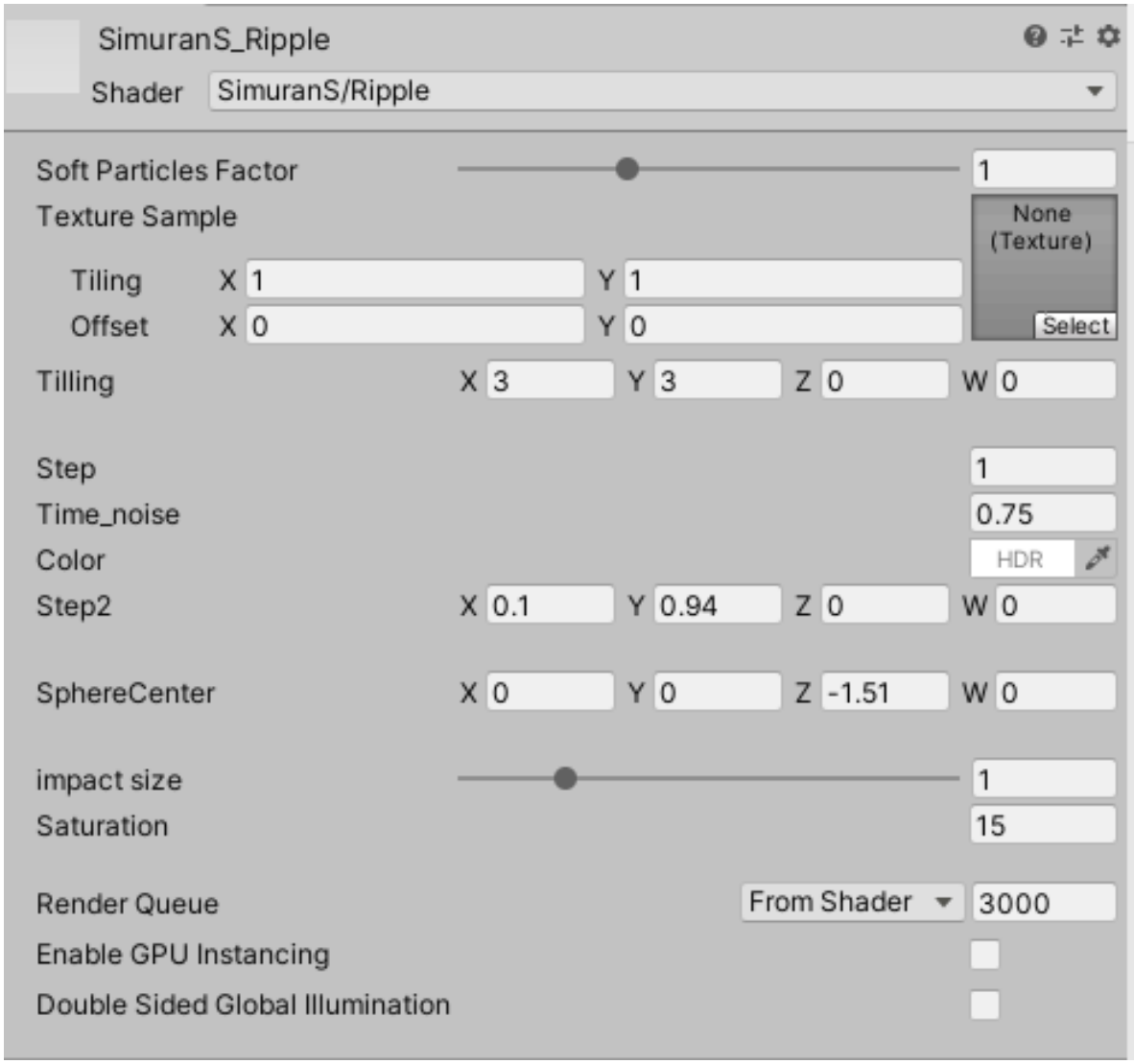


7. Congratulations! Now in the game, when shooting at an object, you can see the Ripple effect that will repeat the contours of this object.

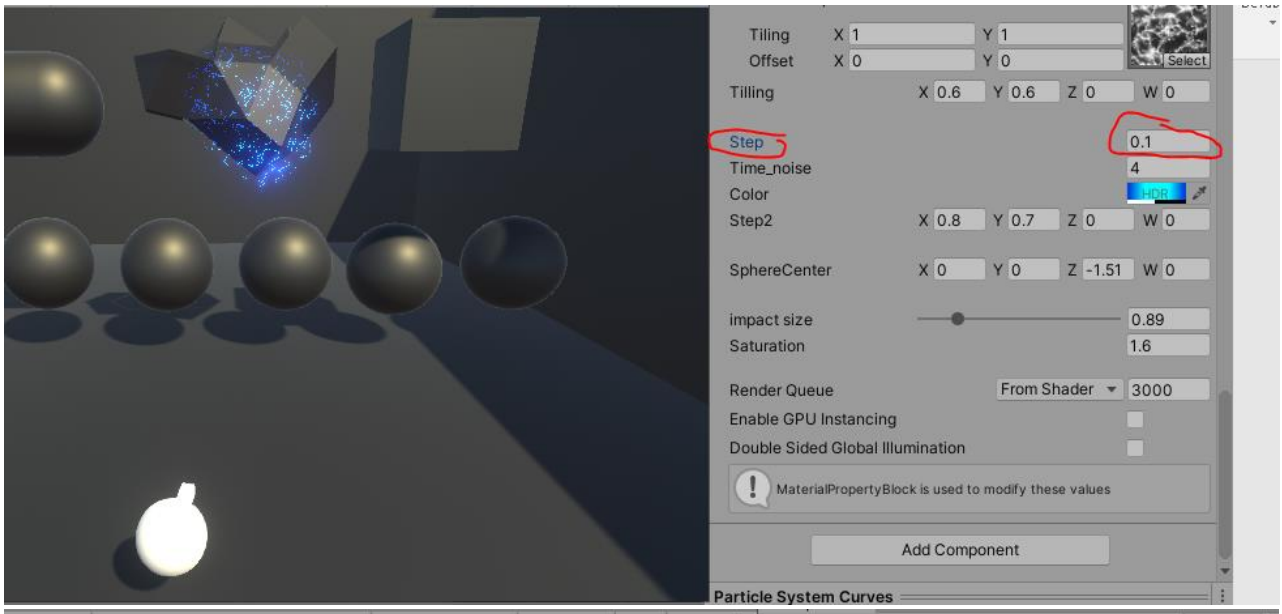


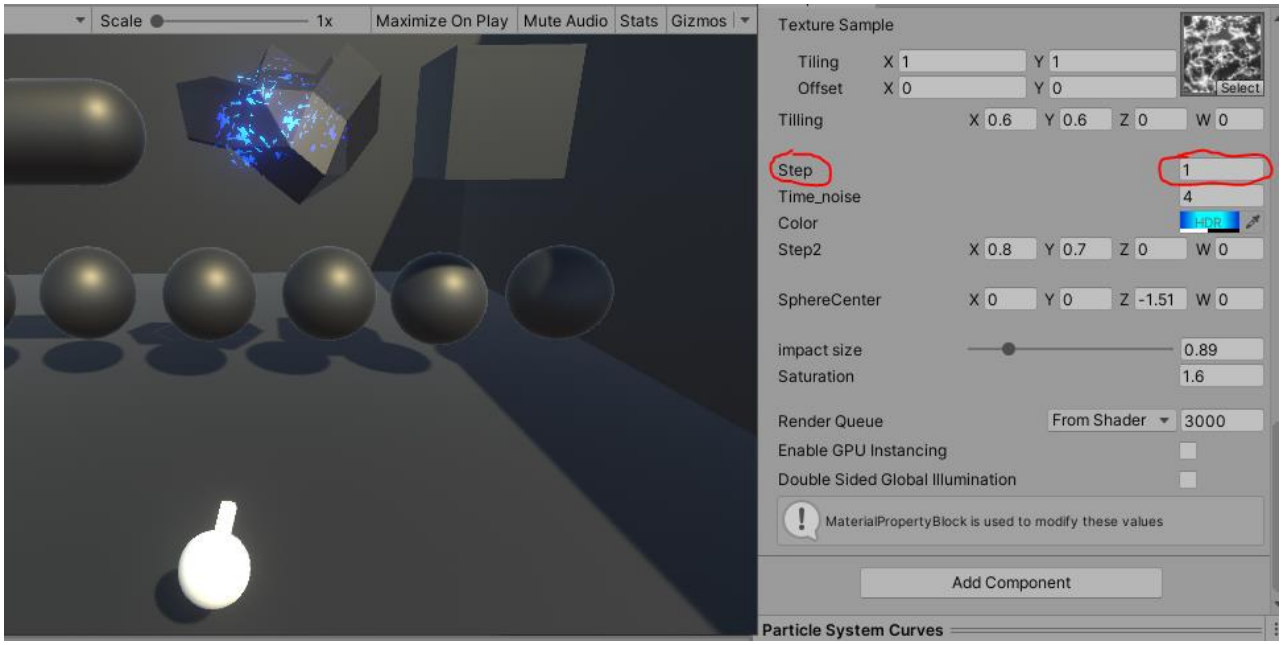
It works with any meshes, but it is advisable to make the corners softer in the model so

that there are no unexpected problems with the display.

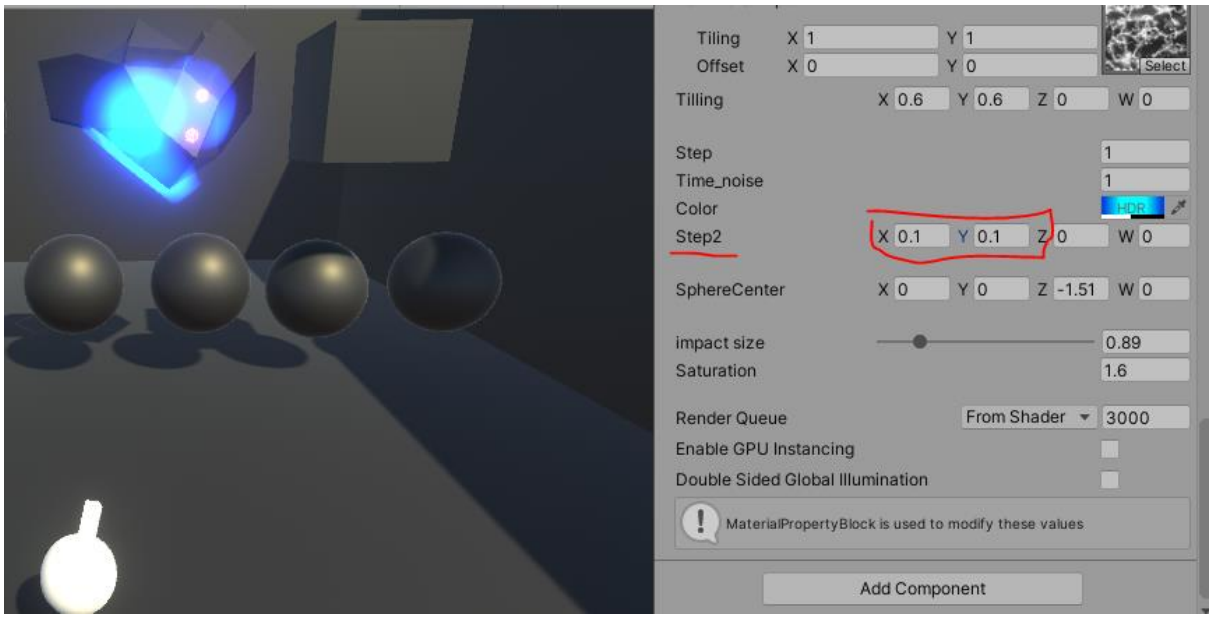


1. Texture Sample – texture of shield.
2. Tilling – scaling the texture.
3. Step – texture display intensity.

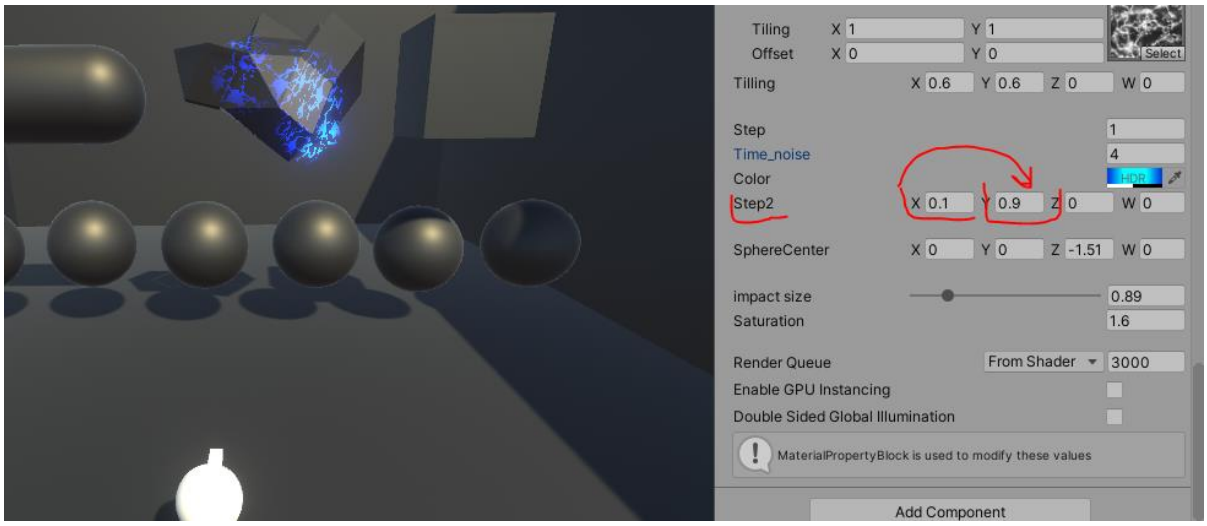




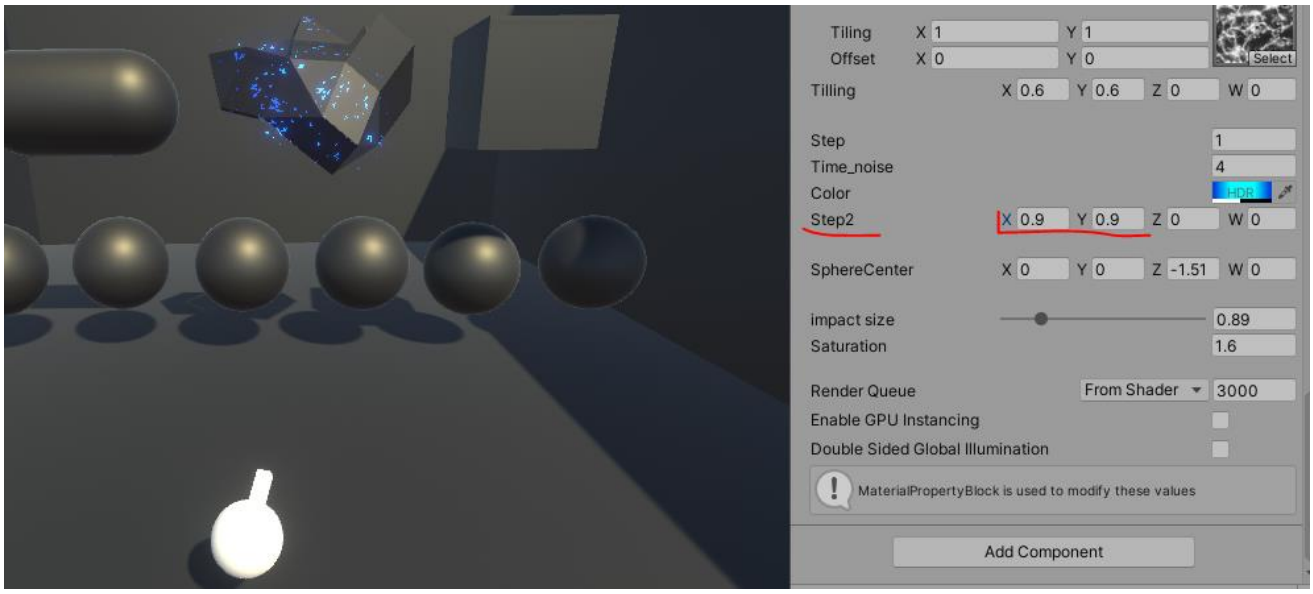
4. Time_noise – texture flicker frequency.
5. Color – texture color.
6. Step2 – The boundaries of the change of the variable Step over time, pulsating from the value of X, to the value of Y.



Without flickering, the minimum values will give very bold lines in the texture.



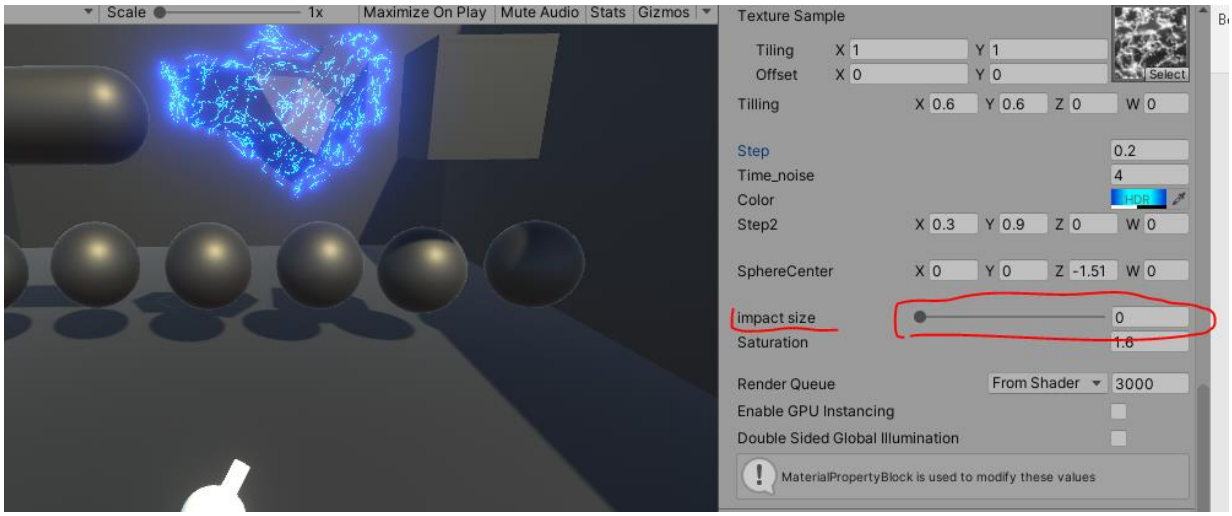
If you make values from 0.1 to 0.9, then the size of the stripes in the texture will pulsate. A flicker will be visible.



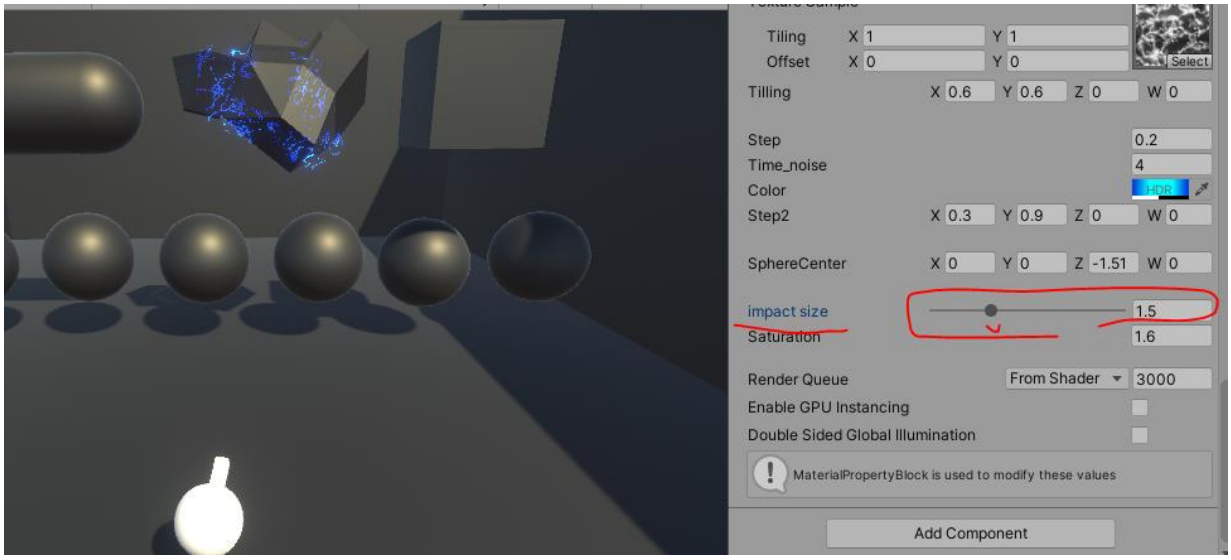
A high parameter will give very little texture visibility. Also, flickering will not occur. To have a flicker, you need to set parameters for example from 0.3 to 0.8 or so.

7. SphereCenter is a service variable. Use it in your script to display impacts if you use other weapons besides the demo cannon provided in the project. (The change does not affect anything; it is just output so that the name for the scripts is visible).

8. Impact size – the size of the mark from the projectile. A higher value gives a smaller footprint spot.

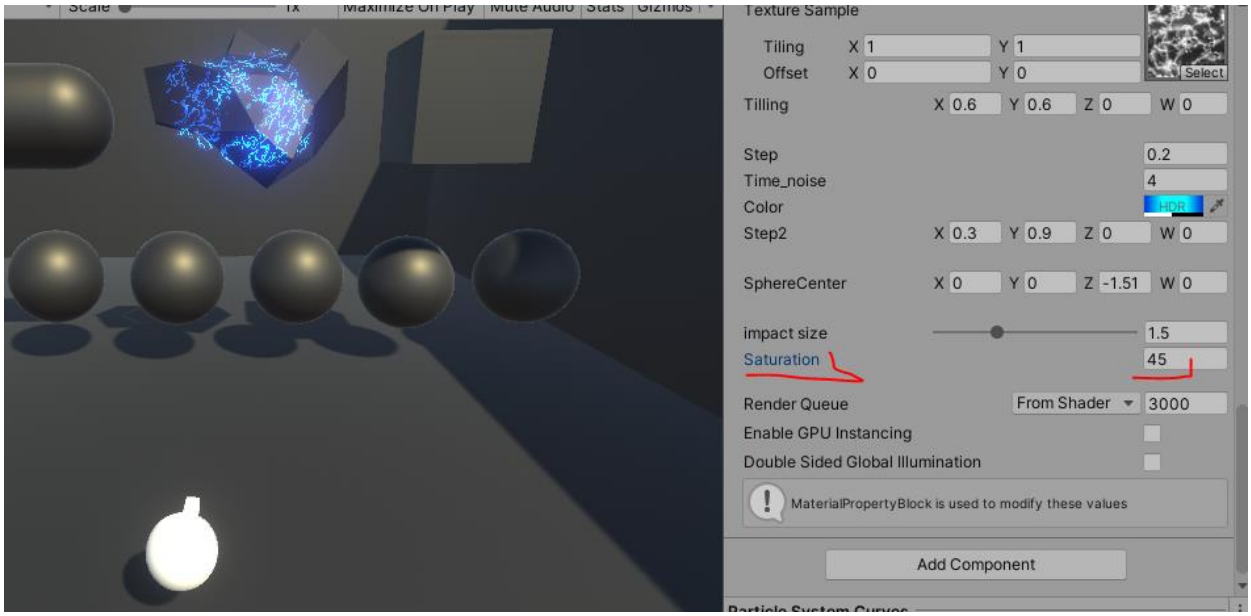


With a value of 0, Ripple will be shown completely capturing the entire object.

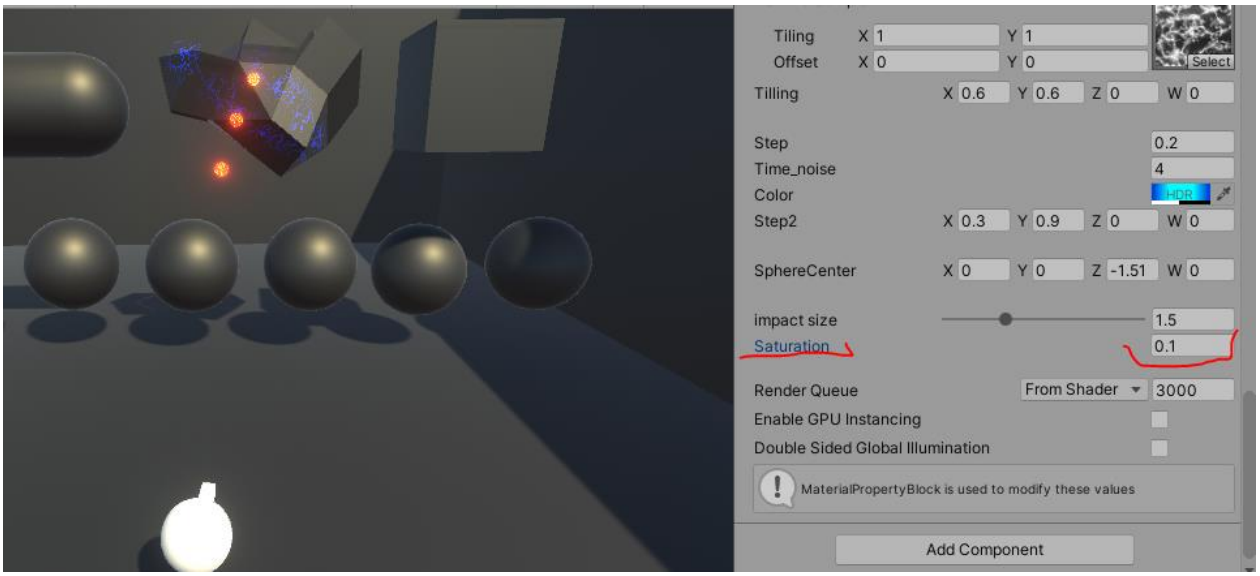


When the value is increased, more of the object will remain hidden.

9. Saturation – sets the color saturation. It is recommended to set values from 0.1 to 45.

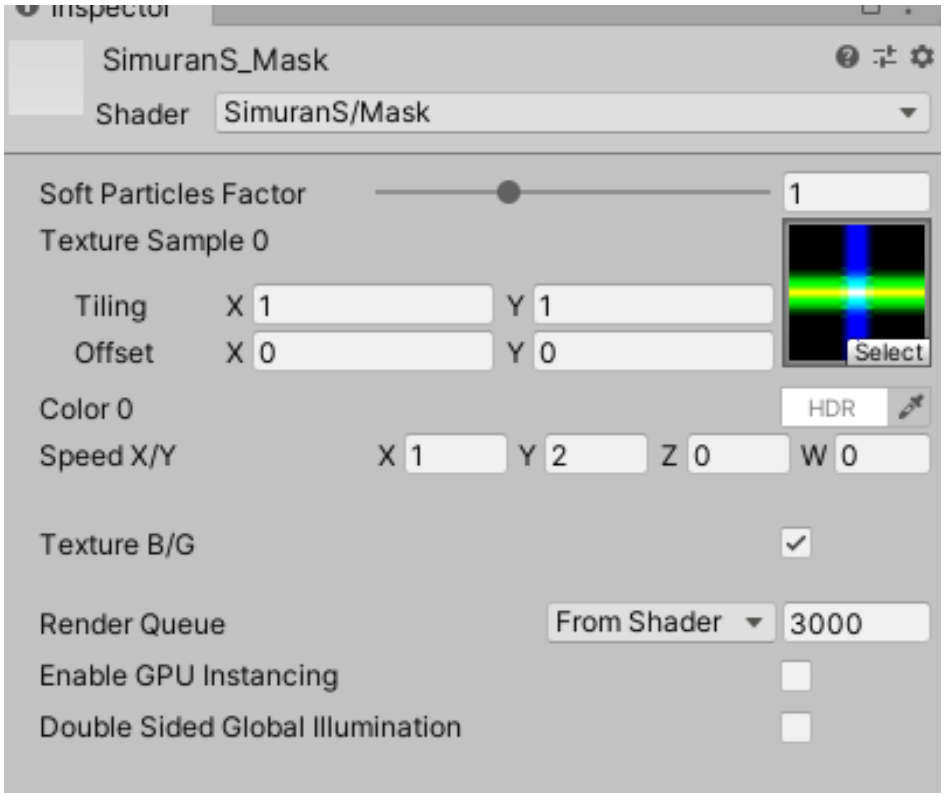


Value 45.



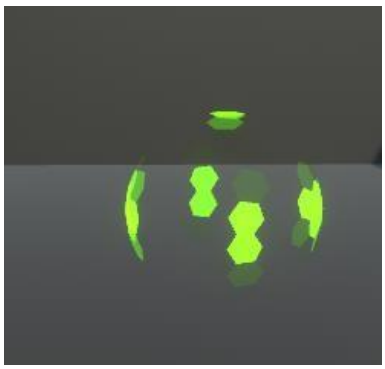
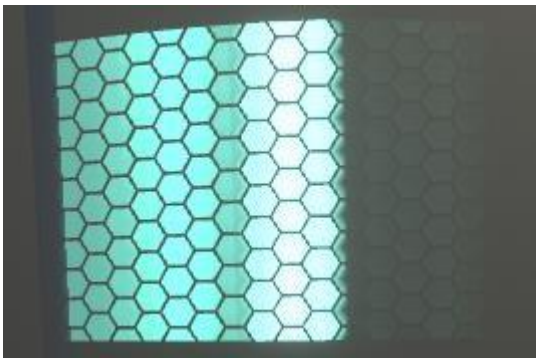
Value 0.1

2.4 Mask (additional shader material).



- It uses 1 texture (2 channels) and an X/Y texture speed.

-Example for spherical and flat objects. Used for flickering.

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